

# Tao Jing (荆韬)

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🔗 astro-jingtao.github.io    🌐 astro-jingtao    🆔 0009-0004-6271-4321

## Research Interests

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- Investigate stellar feedback and its impact on star formation and quenching processes at sub-galactic scales using multi-band imaging and integral field spectroscopy.
- Study the coupling of multiphase interstellar medium.
- Explore the microstructure of shock- or photon-ionized nebulae in the Milky Way and nearby galaxies to refine ionization models and improve hydrodynamic simulations.
- Develop and tailor advanced machine learning techniques for efficient analysis and interpretation of astronomical datasets.

## Education

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- **Doctor of Philosophy**, Astronomy  
Department of Astronomy, Tsinghua University (THU) 2020 - Now
- **Bachelor of Science**, Astronomy  
Department of Astronomy, Xiamen University (XMU) 2016 - 2020

## Publications

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- **Jing, T.** and Li, C. (2025), Correlations of ALMA CO(2-1) with JWST mid-infrared fluxes down to scale of  $\lesssim 100$  parsec in nearby star-forming galaxies from PHANGS. arXiv e-prints, submitted to ApJ, arXiv:2511.10464
- **Jing, T.** and Li, C. (2025), Regression for Astronomical Data with Realistic Distributions, Errors, and Non-linearity. AJ, 170, 45, 10.3847/1538-3881/add891
- **Jing, T.** and Li, C. (2024), On the Origin of Quenched but Gas-rich Regions at Kiloparsec Scales in Nearby Galaxies. ApJ, 975, 17, 10.3847/1538-4357/ad7367
- **Jing, T.**, Li, C., Yan, R., Cheng, C., Zhang, W., Ji, X., Li, N., Wang, J., Wu, C., and Yuan, H. (2024), Criss-cross Nebula: A Case Study of Shock Regions with Resolved Microstructures at Scales of 1000 au. ApJ, 962, 51, 10.3847/1538-4357/ad11d1
- Gao, Y., Liang, S., Tan, Q., Wang, E., Liu, H., Wang, H., **Jing, T.**, Wang, X., Liu, K., Gai, N., Tang, Y., Wang, Y., and Li, Y. (2026), Revisiting the Origin of the Star-Forming Main Sequence Based on a Volume-Limited Sample of 25,000 Galaxies. arXiv e-prints, Universe accepted, arXiv:2602.21500,
- Sui, C., Mao, Y., Zhao, X., **Jing, T.**, and Wandelt, B. D. (2026), How to Evaluate the Sufficiency and Complementarity of Summary Statistics for Cosmic Fields: An Information-theoretic Perspective. ApJ, 998, 329, 10.3847/1538-4357/ae3aa4
- Ma, C., Sun, Z., **Jing, T.**, Cai, Z., Ting, Y.-S., Huang, S., and Li, M. (2026), Can AI Dream of Unseen Galaxies? Conditional Diffusion Model for Galaxy Morphology Augmentation. ApJS, 282, 25, 10.3847/1538-4365/ae1f10
- Guo, R., Li, C., Zhou, S., Li, N., **Jing, T.**, and Cheng, Z. (2025), Mapping Dust Attenuation at Kiloparsec Scales. II. Attenuation Curves from Near-ultraviolet to Near-infrared. Research in Astronomy and Astrophysics, 25, 065017, 10.1088/1674-4527/add673
- Sui, C., Zhao, X., **Jing, T.**, and Mao, Y. (2023), Evaluating Summary Statistics with Mutual Information for Cosmological Inference. Machine Learning for Astrophysics, 30, 10.48550/arXiv.2307.04994

## Collaborations

- [Affordable Multiple Aperture Spectroscopy Explorer \(AMASE\) project](#) [↗](#): a in-processing project that will pair 100 identical multi-fiber spectrographs to map the ionized gas in the Milky Way and nearby galaxies at the spatial resolution of half an arcminute and a spectral resolution of  $R=15,000$ .

I am in Science Team of AMASE and work for the HII region sample selection and construct mock data cubes from simulations. I am also leading a subproject to study microstructures in shocked regions.

## Approved Proposals

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|---------------------------------------------------------------------------------------------------------|-----|
| ◦ PMO Delingha Millimeter Telescope 2022 (30.5 hours awarded)                                           | PI  |
| Title: Detecting $^{12}\text{CO}$ in the Criss-cross Nebula to study shock-molecular cloud collision.   |     |
| ◦ SITELLE@CFHT 2022B (5.5 hours awarded)                                                                | PI  |
| Title: High resolution IFS on Criss-cross Nebula as a case study of microstructure of shock regions     |     |
| ◦ MeerKAT 2022 (7.5 hours awarded)                                                                      | PI  |
| Title: High resolution HI observation on Criss-cross Nebula: the case study for shock-cloud interaction |     |
| ◦ SITELLE@CFHT 2024B (PI: Xiao Li, 5.6 hours awarded)                                                   | CoI |
| Title: Large FoV IFS on UGC1382 as a case study of quenched but HI-rich galaxies                        |     |

## Technical Skills

- **Programming Language:** Python, C, FORTRAN, JavaScript, Mathematica,  $\text{\LaTeX}$
- **Ionization Model:** [MAPPINGS V](#) [↗](#)
- **Spectrum Analysis:** [pPXF](#) [↗](#), self-developed code
- **Simulation Post-processing:** [COLT](#) [↗](#), [CHIMES](#) [↗](#)
- **Machine Learning Packages:** [TensorFlow](#) [↗](#), [JAX](#) [↗](#), [PyTorch](#) [↗](#), [scikit-learn](#) [↗](#), [AutoGluon](#) [↗](#)

## Selected Talks and Posters

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|---------------------------------------------------------------------------------------------|------|
| ◦ The 12th Sino-German Meeting on Galaxy Formation                                          | 2025 |
| Poster + Flash talk, Chengdu, China                                                         |      |
| ◦ Decoding Galactic Evolution through the Interplay of the Multi-Phase Inter-stellar Medium | 2025 |
| Poster + Flash talk, Nagoya, Japan                                                          |      |
| ◦ The 12th Symposium on Interstellar Physics and Chemistry                                  | 2025 |
| Oral talk, Zhangye, China                                                                   |      |
| ◦ Symposium on Evolutionary Laws of Galactic Ecosystems over Billions of Years              | 2025 |
| Poster + Flash talk, Beijing, China                                                         |      |
| ◦ The 27th Guoshoujing Symposium on Galaxies and Cosmology                                  | 2025 |
| Oral talk, Changsha, China                                                                  |      |
| ◦ AMASE Science Meeting                                                                     | 2025 |
| Oral talk, Xiangtan, China                                                                  |      |
| ◦ JWST Discussion Workshop                                                                  | 2025 |
| Invited talk, Given remotely                                                                |      |
| ◦ The Ecosystem of Gas and Dust in Galaxies Near and Far                                    | 2024 |
| Oral talk, Chiang Mai, Thailand                                                             |      |
| ◦ FAST Deep Sky Survey Annual Meeting 2024                                                  | 2024 |
| Oral talk, Beijing, China                                                                   |      |
| ◦ THU Galaxy & Cosmology Seminar                                                            | 2024 |
| Invited talk, Beijing, China                                                                |      |
| ◦ The 11th Symposium on Interstellar Physics and Chemistry                                  | 2024 |
| Oral talk, Nanjing, China                                                                   |      |

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| ◦ AMASE Science Meeting                                    | 2024 |
| Two oral talks, Beijing, China                             |      |
| ◦ The 26th Guoshoujing Symposium on Galaxies and Cosmology | 2024 |
| Oral talk and Poster, Suzhou, China                        |      |
| ◦ Resolving Galaxy Ecosystems Across All Scales 2023       | 2023 |
| Oral talk, Hongkong, China                                 |      |
| ◦ ACAMAR Gas in Galaxies Workshop 2023                     | 2023 |
| Oral talk and Poster, Perth, Australia                     |      |
| ◦ LAMOST MRS-N and Star Formation Evolution Seminar        | 2023 |
| Oral talk, Xuyi, China                                     |      |
| ◦ SDSS Collaboration Meeting 2021                          | 2021 |
| Oral talk, Given remotely                                  |      |
| ◦ The 23th Guoshoujing Symposium on Galaxies and Cosmology | 2021 |
| Oral talk, Hangzhou, China                                 |      |

## Teaching, Professional Services, and Student Supervision

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|----------------------------------------------------------------------------------------------------|-------------|
| ◦ Jian He (何鉴), undergrad at USTB                                                                  | 2026 - now  |
| ◦ Qianyi Liu (刘谦益), undergrad at THU                                                               | 2025 - now  |
| ◦ LOC, Symposium on Evolutionary Laws of Galactic Ecosystems over Billions of Years                | 2025.06     |
| ◦ LOC, AMASE Science Meeting                                                                       | 2024.05     |
| ◦ Organizing Committee of THU Data Science Club/Machine Learning Session                           | 2021 - 2023 |
| ◦ Teaching Assistant, Observational Cosmology: From the Solar System to the Depths of the Universe | 2020 - 2021 |

## Honors and Awards

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|---------------------------------------------------------------------------|-------------|
| ◦ Tsinghua University Comprehensive Excellence Scholarship (First Prize)  | 2024 - 2025 |
| ◦ Tsinghua University Comprehensive Excellence Scholarship (Second Prize) | 2023 - 2024 |
| ◦ AMD Scholarship                                                         | 2020 - 2021 |
| ◦ Shanghai Astronomical Observatory Guangqi Scholarship (First Prize)     | 2018 - 2019 |
| ◦ Xiamen University Tan Chong International Limited Scholarship           | 2017 - 2018 |

## References

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| ◦ Prof. Cheng Li       | Tsinghua University                 |
| cheng.leech@gmail.com  |                                     |
| ◦ Prof. Jing Wang      | Peking University                   |
| jwang_astro@pku.edu.cn |                                     |
| ◦ Prof. Renbin Yan     | The Chinese University of Hong Kong |
| rbyan@cuhk.edu.hk      |                                     |